

What the metrics can and cannot tell us

Paper 6 in a series on UK All-Weather racing outcomes

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Return on investment has been reported throughout this series, alongside the ranking measures, and it cannot carry the model-to-model comparison on its own. Section 1 shows why: of the four consecutive-model differences in single-bet win return, recomputed on one bet rule so they read as a sequence, only the first step is distinguishable from zero, the three later intervals running 22 to 27 points wide against point differences no larger than 10; and the same model under the same rule moves 33 points between two adjacent sets of races. Section 2 keeps both of Owen’s scoring rules, which score different outcomes — P1 the race, P2 the runner-rows as independent binary events — and recomputes them on disjoint bins instead of nested thresholds, against two benchmarks: the over-round-adjusted settled starting price and the overnight racecard forecast price. On 18,411 runner-rows both rules give the same ordering, every model loses to the starting price and beats the forecast price with all twelve intervals excluding zero, and both rules separate the paper-5 encoder from the two earlier models where the return could not. Section 3 fits how much of the model’s disagreement with the forecast price the market later accepts: the pooled coefficient is 0.689 with a standard error of 0.011, so the market’s later move goes about two thirds of the way in the encoder’s direction — though a forecast is a published opinion rather than a traded price, which makes that an upper bound.

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1 Return on investment

Return on investment has been reported throughout this series, alongside the ranking measures, and it cannot carry the model-to-model comparison on its own. Each figure is an unbiased estimate of what that rule returned on the races it was measured on, and nothing was miscomputed. The intervals around the differences between them are simply wider than the differences the series was trying to detect, so a sequence of point estimates across papers cannot establish that the models improved.

Two demonstrations follow. The first is the published record: the four consecutive-model differences and their paired bootstraps. The second holds the model and the rule fixed and changes only the races.

Everything here runs on stored predictions and refits nothing, so these comparisons are out of sample as comparisons. The models being compared are not: they were selected with knowledge of these test races across five papers, where the feature set, the objective, the function class, the encoder architecture and the rung ladder were each chosen after seeing an earlier paper's test results.

1.1 The published record

Single-bet win return, one bet per race under Owen's naive rule (Owen 2019) at the series-standard thresholds, settled at the real starting price:

Table 1: Single-bet win return across the series, on the test split.

Model	Bets	Wins	ROI	ROI less top winner
2a	1,305	151	-25.91%	-27.16%
2b	1,148	137	-15.11%	-16.52%
3	1,133	124	-22.31%	-23.74%
5	1,064	131	-12.19%	-13.52%

Paper 1 is absent because it has no single-bet figure: it reported only the bet-every-qualifier rule, at -28.17% over 1,713 bets.

The point estimates improve by 13.7 points from paper 2a to paper 5, which reads as steady improvement. The paired race-level bootstraps of the consecutive differences do not support that reading.

The four differences below were recomputed for this paper on one bet rule, because the versions the series published do not share one: papers 2b and 3 compared return on the bet-every-qualifier rule while paper 5 compared it on the single-bet rule, and the two rules move the earlier point differences by about three points. The recomputation uses the series' own paired bootstrap and reproduces paper 5's two published figures exactly, on the point difference and both interval ends, so all four rows are the output of one procedure.

Table 2: The four contrasts as the series published them, with the bet rule each was computed on.

Contrast	Basis	Difference	90% interval
2b - 2a	bet every qualifier	+8.02 pp	[+0.97, +14.81]
3 - 2b	bet every qualifier	-4.28 pp	[-13.18, +4.66]
5 - 3	single bet per race	+10.11 pp	[-3.96, +23.34]
5 - hand-built	single bet per race	+7.16 pp	[-4.77, +18.23]

Table 3: The same four contrasts on the single-bet rule. Paired race-level bootstrap, B = 2000, seed 42.

Contrast	Difference	90% interval	Races	Excludes zero
2b - 2a	+10.80 pp	[+1.76, +18.85]	2,193	yes
3 - 2b	-7.64 pp	[-18.57, +3.65]	2,183	no
5 - 3	+10.11 pp	[-3.96, +23.34]	2,183	no
5 - hand-built	+7.16 pp	[-4.77, +18.23]	2,183	no

2b - 2a is the only one of the four whose interval excludes zero, and it is the first step in the sequence rather than the largest difference in it. The three later contrasts all contain zero: their intervals are 22, 27 and 23 points wide against point differences of -7.6 pp, +10.1 pp and +7.2 pp. Every comparison after the first is consistent with an improvement, no change, or a decline.

1.2 One model, one rule, three sets of races

The second demonstration holds the model and the rule fixed and changes only the races. Paper 5's encoder, Owen's rule, one bet per race, nothing refitted and nothing selected.

Table 4: One model and one rule on three sets of races.

Race set	Fit	Races	Bets	Wins	ROI
validation slice, first half	fitting partition	755	391	67	+22.6%
validation slice, second half	fitting partition	750	313	42	-10.2%
test split	full training split	—	1,064	131	-12.2%

The first two rows are the 1,505 races carved out of the training split at 2010-12-15 for paper 5’s model selection, split chronologically at 2012-01-15 into 755 and 750 races. The same model under the same rule returns +22.6% on one and -10.2% on the other, a spread of 33 points across two adjacent sets of races of almost equal size. On the test split it returns -12.2%.

Those three figures are not identical in provenance: the validation slice is scored by the encoder’s fitting-partition fit and the test split by the full-training-split refit, which are two fits of one architecture. The spread between the two halves needs no such qualification, since both are scored by the same fit.

2 Owen’s scoring rules

Paper 1 reports two scoring rules over the runner-rows whose probability exceeds a threshold t : $P_1(t)$, the geometric mean of the probability assigned to winners, and $P_2(t)$, the geometric mean of the per-row Bernoulli likelihood. Both are kept here, with two changes.

2.1 What each rule scores

Every source in this paper is a distribution over the race. The model probabilities come from a softmax or a conditional logit over the race’s runners, and both price columns are renormalised within race, so each sums to one over the field.

Under that constraint the two rules score two different outcomes. **P1 at $t = 0$ is the exponentiated mean, over races, of the log probability assigned to the horse that won** — there is exactly one winner per race, so the mean over winning rows is a mean over races. That is the log score for the race-level categorical outcome. **P2 sums a Bernoulli log-likelihood over every runner-row and divides by the row count**, so it scores each runner’s marginal win probability as an independent binary event and does not use the one-winner constraint. Each is proper for the outcome it scores.

The series’ own headline ranking measure is built the same way as P1: `score_p1_rank` is the exponentiated mean log probability of the observed top-three finishing order, which is P1’s construction at depth three.

2.2 Two changes

Disjoint bins instead of nested thresholds. Subsetting rows on the source’s own probability before scoring affects both rules equally: under $P_1(t)$ and $P_2(t)$ the subset at $t = 0.15$ still contains every row above 0.15, so consecutive thresholds are nearly the same rows and no band can be read on its own. Disjoint bins are what address that. The bins below 0.10 are two, two, two and four points wide, finer than any threshold the series has used, because that is where the models’ long-priced statements sit.

Two benchmarks. The over-round-adjusted settled starting price, as the series has always used, and the overnight racecard forecast price from `daily_runners`, renormalised per race the same way. Paper 4 found the model ties the first and beats the second, so one benchmark alone cannot say whether a model’s probabilities are good.

Every source is scored on the same rows. Paper 4’s price panel supplies both benchmark probabilities and papers 2b and 3’s stored predictions; paper 5’s encoder is joined on from its own stored test predictions.

Table 5: The test split against the rows both benchmarks cover.

Rows	Races	Rows scored	Races scored
18,419	2,183	18,411	2,182

Requiring a usable pre-race forecast price costs 8 rows in 1 race, so both benchmarks are scored on identical rows.

2.3 The three models under both benchmarks

Table 6: Both rules, every source, over every scored runner-row. Higher is better. Race-level bootstrap, B = 2000, seed 42.

Source	P1	P1 interval	P2	P2 interval
Starting price	0.16651	[0.16247, 0.17041]	0.72855	[0.72602, 0.73134]
Paper 5 encoder	0.14732	[0.14464, 0.14997]	0.71594	[0.71386, 0.71807]
Paper 2b logit	0.14290	[0.14027, 0.14546]	0.71298	[0.71093, 0.71516]
Paper 3 GBT	0.14278	[0.14016, 0.14534]	0.71290	[0.71088, 0.71507]
Forecast price	0.14042	[0.13746, 0.14323]	0.71130	[0.70899, 0.71363]

The two rules give the same ordering, Starting price > Paper 5 encoder > Paper 2b logit > Paper 3 GBT > Forecast price, so nothing below turns on which is read.

Table 7: Each model against each benchmark, on both rules, paired by race. Every interval excludes zero.

Rule	Model	Benchmark	Difference	90% interval
P1	Paper 2b logit	Forecast price	+0.00248	[+0.00066, +0.00429]
P1	Paper 3 GBT	Forecast price	+0.00236	[+0.00043, +0.00436]
P1	Paper 5 encoder	Forecast price	+0.00690	[+0.00501, +0.00871]
P1	Paper 2b logit	Starting price	-0.02362	[-0.02630, -0.02094]
P1	Paper 3 GBT	Starting price	-0.02374	[-0.02648, -0.02098]
P1	Paper 5 encoder	Starting price	-0.01920	[-0.02179, -0.01672]
P2	Paper 2b logit	Forecast price	+0.00168	[+0.00040, +0.00298]
P2	Paper 3 GBT	Forecast price	+0.00160	[+0.00022, +0.00307]
P2	Paper 5 encoder	Forecast price	+0.00463	[+0.00330, +0.00592]
P2	Paper 2b logit	Starting price	-0.01557	[-0.01724, -0.01387]
P2	Paper 3 GBT	Starting price	-0.01565	[-0.01734, -0.01391]
P2	Paper 5 encoder	Starting price	-0.01261	[-0.01423, -0.01104]

All 12 contrasts exclude zero: every model loses to the starting price and beats the forecast price on both rules. That is paper 4’s finding in a different measure, and it is why one benchmark is not enough. Against the settled price alone the models look worse than the market; against the overnight price alone they look better; both are true.

2.4 The three models against each other

Table 8: The three models against each other, on both rules, paired by race.

Rule	Contrast	Difference	90% interval	Excludes zero
P1	Paper 2b logit - Paper 3 GBT	+0.00012	[-0.00079, +0.00102]	no
P1	Paper 5 encoder - Paper 2b logit	+0.00442	[+0.00325, +0.00555]	yes
P1	Paper 5 encoder - Paper 3 GBT	+0.00454	[+0.00327, +0.00578]	yes
P2	Paper 2b logit - Paper 3 GBT	+0.00008	[-0.00059, +0.00073]	no
P2	Paper 5 encoder - Paper 2b logit	+0.00296	[+0.00215, +0.00373]	yes
P2	Paper 5 encoder - Paper 3 GBT	+0.00304	[+0.00215, +0.00388]	yes

On both rules the encoder separates from the two earlier models and papers 2b and 3 do not separate from each other. Section 1 could do neither: the single-bet return difference between the encoder and paper 3 was +10.1 pp with an interval 27 points wide. Both rules are computed over all 18,411 runner-rows; the return difference was computed over the races Owen’s rule selects, and within those over the handful that won.

2.5 By bin

The bins carry calibration and no score. A score computed inside a bin is largely set by the bin’s win rate, and P1 reads only the winning rows, so four of the fifty source-by-bin cells hold no winner at all. What a source states against what happens is the reading that survives, and it is the only one the bins are used for here.

Table 9: Observed win rate less mean stated probability, in percentage points. A negative gap means the source states more than it wins. Columns: 2b = paper 2b logit, 3 = paper 3 GBT, 5 = paper 5 encoder, SP = starting price, FP = forecast price.

Bin	2b	3	5	SP	FP
0-2%	-1.5	-1.6	-1.6	-1.1	-1.6
2-4%	-1.7	-1.5	-1.2	-1.1	-1.0
4-6%	-1.9	-1.6	-1.8	-0.6	+0.0
6-10%	-0.1	-0.3	-0.9	-0.3	+0.7
10-15%	-0.2	+0.4	+0.0	+0.2	+0.4
15-20%	+0.4	+0.4	+0.8	+0.4	-0.5
20-25%	+2.5	+0.4	+1.6	+0.4	-1.1
25-30%	+2.1	+2.4	+4.9	+0.7	-2.4
30-35%	+6.9	+4.3	+9.1	+0.8	-1.0
35%+	+2.8	+5.7	+7.9	+6.2	+1.5

Every source states more than it wins at the long-priced end and less than it wins at the short-priced end, and the models overshoot further than the prices do. The encoder’s 30-35% bin wins +9.1 points more often than it says, against +0.8 for the starting price, which is within 1.1 points of its stated probability in every bin but its top one. The nested form could not show this, because it pooled everything above each threshold and the upper bands were swamped by the rows below them.

The bottom bin is thin and its gap is close to minus the stated probability by construction: it holds 19 to 613 rows with at most 2 winners depending on the source.

3 How much of the model’s disagreement the market accepts

Two things bound what follows. The forecast price is a compiler’s published opinion rather than a price money has traded at, and an early exchange price would already contain much of what a compiler misses, so anything measured here is an upper bound on anything tradeable. And a coefficient near zero would close the idea off, while a positive one does not establish it.

Paper 4 fitted a two-stage conditional logit with a market probability and a model probability, both coefficients free, and found the model adds nothing distinguishable against the settled starting price and a great deal against the overnight racecard forecast price.

Table 10: Paper 4’s model-probability coefficients, read from its own stored fits.

Benchmark	Sample	Model	Coefficient	95% interval
settled starting price	pooled	paper 2b logit	0.068	[-0.067, +0.204]
settled starting price	pooled	paper 3 GBT	0.039	[-0.097, +0.175]
overnight forecast price	test-A	paper 2b logit	1.037	[+0.835, +1.238]
overnight forecast price	test-A	paper 3 GBT	1.026	[+0.833, +1.219]
overnight forecast price	test-B	paper 2b logit	0.746	[+0.547, +0.945]
overnight forecast price	test-B	paper 3 GBT	0.723	[+0.539, +0.908]

That result is taken as given and not re-argued. What it implies is that the model’s information, such as it is, sits between the overnight price and the off, which makes the quantity of interest how much of the model’s disagreement with the overnight price the market subsequently accepts.

3.1 The regression

On the test split, on the encoder’s stored predictions, with p_{close} the over-round-adjusted settled starting price, p_{fcst} the renormalised overnight forecast price and $\hat{p}$ the encoder’s probability:

$$\text{logit}(p_{\text{close}}) - \text{logit}(p_{\text{fcst}}) = \alpha + \beta [\text{logit}(\hat{p}) - \text{logit}(p_{\text{fcst}})] + \varepsilon.$$

Standard errors are a **race-level bootstrap**, $B = 2000$, seed 42, which is the resampling unit and replicate count used for every other interval in this paper. The alternative was race-clustered analytic errors; the bootstrap is used for consistency with the rest of the series.

The rows are the same 18,411 runner-rows in 2,182 races that Section 2 scores, because the regression needs both prices and the forecast-price requirement bites in the same place.

Table 11: The pooled fit. Standard errors and intervals from a race-level bootstrap, $B = 2000$, seed 42.

Term	Estimate	Std. error	90% interval
intercept	-0.1338	0.0020	[-0.1371, -0.1303]
slope	0.6892	0.0110	[+0.6709, +0.7072]

The pooled coefficient is 0.689 with a standard error of 0.011 and a 90% interval of [+0.671, +0.707].

What that says is narrow. Where the encoder disagrees with the published forecast, the market's own later move away from that forecast goes about 69% of the way in the same direction, so the encoder anticipates most of a move the market was going to make anyway. It is not a return, not an edge, and not a price improvement. It is agreement between two opinions about a third, and the third is published rather than traded.

The intercept is -0.134. Both price columns are renormalised to sum to one within each race, so no difference in over-round between the two books survives into the fit; what remains is that the settled book is more spread out than the forecast book, putting 30.1% on its favourite against 25.6%, so the typical runner's closing logit sits below its forecast logit even where the model and the forecast agree.

Only the pooled coefficient is reported. There is no breakdown by price band, field size or anything else, because that would be a search and Section 1 is the reason searches on this data do not work.

Owen, Alun. 2019. "Statistical Models of Horse Racing Outcomes Using R." *MathSport International 2019 Conference Proceedings* (Athens, Greece).